

Inequations

Inequations, also known as inequalities, represent relationships between two quantities where they are not necessarily equal. They use symbols such as $>$, $<$, $\geq$, and $\leq$ to indicate whether one quantity is greater than, less than, greater than or equal to, or less than or equal to the other. Inequations can involve linear expressions, indicating a range of possible solutions, or more complex forms like absolute value inequalities, requiring consideration of multiple cases to capture all possible solutions. When solving inequalities, the focus is on isolating the variable while maintaining the relationship denoted by the inequality symbol, with special care taken when multiplying or dividing by negative numbers. Inequations are key in algebra, calculus, and optimization, providing a way to describe conditions where equality does not necessarily hold.

Example 1: Solve $x^2 - 9 > 0$

To find the solution to the inequality $x^2 - 9 > 0$, first identify the critical points where the expression equals zero.

$$1. \ x^2 - 9 = 0 \Rightarrow x^2 = 9 \Rightarrow x = -3 \text{ and } x = 3$$

The points $x = -3$ and $x = 3$ divide the real number line into intervals.

Next, you can test the sign of the expression $x^2 - 9$ in each interval to determine where it is positive/non-negative (> 0).

When $x < -3$: In this interval, both x^2 and -9 are negative. Therefore, $x^2 - 9$ is positive.

When $-3 < x < 3$: Here, x^2 , the factor $(x + 3)$ is positive and the factor $(x - 3)$ is negative, so their product is negative. The interval between -3 and 3 does not satisfy the condition.

When $x > 3$: Both x^2 and -9 are positive. Therefore, $x^2 - 9$ is positive.

$-\infty$	$\dots$	-3	$\dots$	0	$\dots$	3	$\dots$	$+\infty$
> 0		$= 0$	< 0		$= 0$	> 0		
$\uparrow$						$\uparrow$		

Based on these analyses, the solution to the inequality $x^2 - 9 > 0$ is

$x \in (-\infty, -3 \cup 3, +\infty)$. So, the solution is all real numbers x such that x is less than -3 or x is greater than 3 .

Example 2: Solve $5 + \frac{(x-2) \cdot (x+2)}{4} > 3 - x$

To solve the inequality $5 + \frac{(x-2) \cdot (x+2)}{4} > 3 - x$, let's first simplify the expression step by step.

Expanding the expression inside the brackets:

$$5 + \frac{(x-2) \cdot (x+2)}{4} > 3 - x$$

Now, let's simplify the expression:

$$5 + \frac{x^2 - 4}{4} > 3 - x$$

We can further simplify the expression:

$$\begin{aligned} 5 + \frac{x^2 - 4}{4} + x &> 3 \\ 5 + \frac{x^2 - 4}{4} + x - 3 &> 0 \\ \frac{x^2 - 4}{4} + x + 2 &> 0 \end{aligned}$$

Now, let's find the common denominator:

$$\begin{aligned} \frac{x^2 - 4 + 4x + 8}{4} &> 0 \\ \frac{x^2 + 4x + 4}{4} &> 0 \end{aligned}$$

Next, let's factor the numerator:

$$\frac{(x+2)^2}{4} > 0$$

Since $(x+2)^2$ is always non-negative for all real values of x , the inequality holds true for all $x \in \mathbb{R}$.

Therefore, the solution to the inequality $5 + \frac{(x-2) \cdot (x+2)}{4} > 3 - x$ is $x \in \mathbb{R}$, excluding $x = -2$, because when $x = -2$, the expression becomes equal to 0, not a number greater than 0.

$$\begin{array}{c|c|c} -\infty & \dots & -2 \\ \hline > 0 & & = 0 \\ \hline \uparrow & & \uparrow \end{array}$$

Thus, the correct solution is $x \in \mathbb{R} \setminus \{-2\}$, indicating that all real numbers except -2 satisfy the inequality.